Student ID :

Student Name :

Total ⇨ 100

1	2	3	4	5	6	7	8	9	10
6	*4*	*8*	*6*	*24*	*8*	*4*	*6*	*16*	*18*

Marmara University – Faculty of Engineering

Computer Engineering

Spring 2020
CSE1142 Computer Programming II

Online Homework #1

01 June 2020

I hereby swear that the work done on this online homework is totally my own; and on my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for this homework. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

Full Name: *Signature:*

Solution and Submission Notes:

1) Textbooks, slides and notes are closed.
2) Internet usage, computers and cell phones are not allowed.
3) This online homework consists of 10 questions and 14 pages.
4) Please write your answers clearly and neatly to the appropriate place.
5) Show all your work.
6) Write your student ID, first name and last name on top of each solution page.
7) If you have not printed this PDF file (14 pages), write the sentences above in the frame on top of the first page and the date in your handwriting, and sign it.
8) If you have not printed this PDF file (14 pages), solve all the questions in your handwriting on blank A4 sheets.
9) Scan all the solution pages to a single PDF file, name it as “FirstName_LastName.pdf”.
10) Upload your PDF file to http://ues.marmara.edu.tr before deadline.

1) [6 pts] Trace the code below and write both the line numbers and the output.

```
 1: public class Test {
 2:   public static void main(String[] args) {
 3:     new Car();
 4:   }
 5: }
 6:
 7:
 8: abstract class Vehicle {
 9:   protected Vehicle() {
10:     System.out.print("V0 ");
11:   }
12:
13:   protected Vehicle(double weight, String color) {
14:     System.out.print("V2 ");
15:   }
16: }
17:
18:
19: class Car extends Vehicle {
20:   public Car() {
21:     this(4);
22:     System.out.print("C0 ");
23:   }
24:
25:   public Car(int noOfSeats) {
26:     this(noOfSeats, 1250.0, "red");
27:     System.out.print("C1 ");
28:   }
29:
30:   public Car(int noOfSeats, double weight, String color) {
31:     super(weight, color);
32:     System.out.print("C3 ");
33:   }
34: }
```

Line numbers ⇨

Output ⇨

2) [4 pts] What is in the ArrayList list after the following code is executed?

```
import java.util.*;
public class Test {
  public static void main(String[] args) {
    ArrayList<Integer> list = new ArrayList<Integer>();
    list.add(7);
    list.add(2);
    list.add(1);
    list.add(5);
    list.add(0,3);
    list.remove(3);
  }
}
```

list is ⇨

3) [8 pts -> 2 pts each] Suppose we have the following classes and methods:

```
class Spr {
  void f() { System.out.println("Spr.f()"); }
  void f(int n) { System.out.println("Spr.f(int)"); }
  void g(int n) { System.out.println("Spr.g(int)"); }
}

class Sub1 extends Spr {
  void f() { System.out.println("Sub1.f()"); }
  void h() { System.out.println("Sub1.h()"); }
}

class Sub2 extends Sub1 {
  void g(int n) { System.out.println("Sub2.g(int)"); }
  void g(String s) { System.out.println("Sub2.g(String)"); }
  void h(int n) { System.out.println("Sub2.h(int)"); }
}
```

For each of the following statements; what is the the output? If there is an error (compile-time or runtime type), explain the reason. Consider each part separately.

a)
```
Spr a = new Sub2();
a.f();
```

b)
```
Spr a = new Sub2();
a.g(10);
```

c)
```
Spr a = new Sub2();
a.g("x");
```

d)
```
Sub1 b = new Sub1();
Sub2 c = b;
c.h(17);
```

4) [6 pts] What is the output of the following program?

```
class SuperClass {
  public void getInfo() {
    System.out.println("SuperClass.getInfo");
  }
  public static void delete() {
    System.out.println("SuperClass.delete");
  }
}

class SubClass extends SuperClass {
  public static void delete() {
    System.out.println("SubClass.delete");
  }
  public void getInfo() {
    System.out.println("SubClass.getInfo");
  }
}

public class Test {
  public static void main(String args[ ])  {
    SuperClass a = new SubClass();
    a.delete();
    a.getInfo();
  }
}
```

Output ⇨

5) [24 pts -> 3 pts each] Assume that all the code below (InfX, ClsA, ClsB, SubC) compiles and runs without error. Assume that you have independent blocks of code that appear in a Test class. For each block of code, state the execution result (compile error, runtime error - ClassCastException, or no problem) for the following statements and explain the reason for the error cases. If there is no problem, write down the output of the program. Consider each case seperately.

```
public interface InfX {
  public void display();
}
```

```
public class ClsA implements InfX {
  public void display() {
    System.out.println("ClsA.display");
  }
  public void click() {
    System.out.println("ClsA.click");
  }
}
```

```
public class ClsB implements InfX {
  public void display() {
    System.out.println("ClsB.display");
  }
  public void click() {
    System.out.println("ClsB.click");
  }
  public void display(Object o) {
    System.out.println("ClsB.display.object");
  }
}
```

```
public class SubC extends ClsB {
  public double value1;
  private double value2;
  public SubC() {
    value1 = 0;
    value2 = 0;
  }
  public SubC(double x, double y){
    value1 = x;
    value2 = x;
  }
  public void display() {
    System.out.println("SubC.display");
  }
  public static void display(double value1, double value2) {
    System.out.println("SubC.display.d2");
  }
  private void display(int x) {
    System.out.println("SubC.display.int");
  }
  public void display(double y) {
    System.out.println("SubC.display.d1");
  }
  public double getValue1() {
    return value1;
  }
  public double getValue2() {
    return value2;
  }
}
```

a)
```
SubC a = new SubC(3.0,2.0);
a.display();
```

b)
```
SubC b = new SubC(3.0,2.0);
SubC c = new SubC();
c.display(1 + " ");
```

c)
```
InfX d = new InfX();
d.display();
```

d)
```
InfX e = new ClsA();
e.click();
```

e)
```
InfX f = new SubC();
ClsB g = (SubC) f;
g.display();
```

f)
```
InfX h = new SubC();
ClsA k = (ClsA) h;
k.display();
```

g)
```
InfX m = new SubC();
((SubC)m).display(1.0,2.0);
```

h)
```
ClsA n = new ClsA();
InfX p = new SubC();
n = (SubC) p;
n.display();
```

6) [8 pts] What is the output of the following program?

```
class Test {
  public static void main(String[] args) {
    try {
      aMethod();
      System.out.println("main.aMethod");
    }
    catch (RuntimeException ex) {
      System.out.println("main.RunEx");
    }
    catch (Exception ex) {
      System.out.println("main.Ex");
    }
    finally {
      System.out.println("main.Fin");
    }
  }
  static void aMethod() throws Exception {
    try {
      String s = "11.42";
      Integer.parseInt(s); // Cause a NumberFormatException
      int i = 0;
      int y = 2 / i;
      System.out.println("aMethod.Parse");
    }
    catch (NumberFormatException ex) {
      System.out.println("aMethod.NumForEx");
    }
    catch (RuntimeException ex) {
      System.out.println("aMethod.RunEx");
    }
    finally {
      System.out.println("aMethod.Fin");
    }
  }
}
```

7) [4 pts] What is the output of the following program?

```
public class Test {
  public static void main(String[] args) throws Exception {
    try {
      int m = 0;
      int k = 5 / m;
    }
    catch (Exception e) {
      System.out.println("Exception");
    }
    catch (ArithmeticException ae) {
      System.out.println("Arithmetic Exception");
    }
    System.out.println("Finished");
  }
}
```

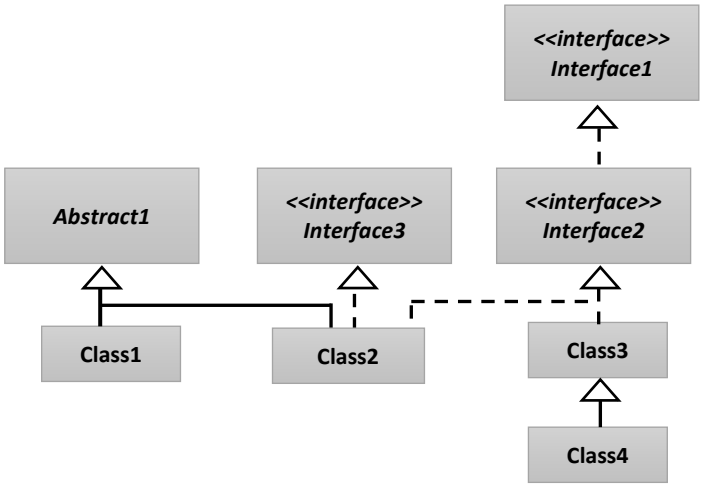
8) [6 pts] What is the output of the following program?

```
public class Demo {
  String str = "X ";
  void A() throws Exception {
    try {
      str +="A ";
      try {
        str +="Y ";
        try {
          str +="D ";
          throw new Exception();
        }
        catch(Exception ex) {
          str +="Z ";
          throw new Exception();
        }
      }
      catch(Exception ex) {
        str +="K ";
      }
    }
    catch(Exception ex){
      str +="F ";
    }
  }

  void display() {
    System.out.println(str);
  }

  public static void main(String[] args) throws Exception {
    Test a = new Test();
    a.A();
    a.display();
  }
}
```

9) [16 pts] Answer the following questions using the figure and the code below.




```
Interface1 obj1 = new Class3();
Abstract1 obj2 = new Class1();
Object obj3 = new Class2();
Class3 obj4 = new Class4();
```

a) [1 pt] Is obj1 instanceof Interface2? ❍ True ❍ False

b) [1 pt] Is obj3 instanceof Interface1? ❍ True ❍ False

c) [1 pt] Is obj4 instanceof Interface3? ❍ True ❍ False

d) [1 pt] Is obj2 instanceof Interface1? ❍ True ❍ False

e) [2 pts each] State the execution result (compile error, runtime error - ClassCastException, or no problem) for the following statements and explain the reason for the error cases. Each case is independent from each other. Suppose the method **aMethod** is defined in the **Class4**.

i) `obj1.aMethod();`

ii) `Interface3 obj5 = obj2;`

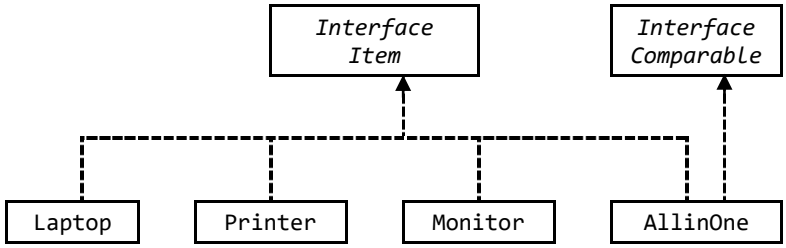
iii) `((Class3)obj3).aMethod();`

iv) `((Class4)obj4).aMethod();`

f) [4 pts] Suppose that method "**`void m2(){};`**" is in **Abstract1**. Which classes have to implement it?

10) [18 pts] An online shopping site offers a variety of laptops, printers, and monitors. A customer may also buy all in one (AllinOne) including these three. The price of AllinOne is the sum of the two highest-priced items in the AllinOne, *i.e.* one item with the lowest price is free. Each item has a name and a price. The four types of the items are represented by the four classes **Laptop**, **Printer**, **Monitor**, and **AllinOne**. All four classes implement the following ***Item*** interface.

The following diagram shows the relationship between the Interface ***Item*** and the classes **Laptop**, **Printer**, **Monitor**, and **AllinOne**.



For example, assume that items sold on the site are as follows. The objects listed under each heading are instances of the class indicated by the heading.

Laptop	Printer	Monitor
"Apple MacBook Pro" 2000.00	"Canon LBP6310" 150.75	"Apple ThunderBolt" 1800.00
"Dell XPS" 1500.00	"HP Deskjet 5820" 120.00	"Dell Curve GT" 900.00

- Customers may create **AllinOne** items, each of which includes a laptop, a printer, and a monitor.
- The name of the **AllinOne** is composed of the names of the laptop, the printer, and the monitor, in order, each separated by "/" and followed by a space and then "**AllinOne**".
- The price of the **AllinOne** is the sum of the two highest-priced items in the AllinOne; *i.e.* one with the lowest price is free.
- An AllinOne consisting of an Apple MacBook Pro, a Canon LBP610, and an Apple ThuderBolt will have the name "**Apple MacBook Pro/Canon LBP610/Apple ThuderBolt AllinOne**" and a price of **$3800.00** (the two highest prices are $2000.00 and $1800.00). Similarly, an AllinOne consisting of a Dell XPS, an HP Deskjet 5820, and a Dell Curve GT will have the name "**Dell XPS/HP Deskjet 5820/Dell Curve GT AllinOne**" and a price of **$2400.00** (the two highest prices are $1500.00 and $900.00).

Write the class **AllinOne** that implements the interfaces ***Item*** and ***Comparable***. Your implementation must include a constructor that takes three parameters representing a laptop, a printer, and a monitor. The main method is given to test the result.

```
interface Item {
  String getName();
  double getPrice();
}
```

```
class SimpleItem implements Item {
  private String name;
  private double price;
  public SimpleItem(String aName, double aPrice){
    name = aName;
    price = aPrice;
  }
  public String getName() { return name; }
  public double getPrice() { return price; }
  public String toString () {
    return (getName() + " $" + getPrice());
  }
}
```

```
class Monitor extends SimpleItem{
  public Monitor(String name, double price) {
    super(name, price);
  }
}

class Printer extends SimpleItem {
  public Printer(String name, double price) {
    super(name, price);
  }
}

class Laptop extends SimpleItem {
  public Laptop(String name, double price) {
    super(name, price);
  }
}

/*
- Declare the calss AllinOne. It must implement the Item and Comparable interfaces.
- Declare the fields that you need for an AllinOne object.
- Write a constructor that takes a Laptop, a Printer, and a Monitor, in order.
- Write the method getName to return laptop name/printer name/monitor name AllinOne.
- Write the method getPrice to return the sum of the prices of two highest-price items in the AllinOne.
- Write the method compareTo and return 1, if the price of the AllinOne is greater than the price of the argument; 0, if they are equal; or -1, if the price of the AllinOne is less than the price of the argument.
*/

// This is a main method to test your code.
// You do not need to write down the output of the main method.
public static void main(String[] args) {
  Laptop mac = new Laptop("Apple MacBook Pro ", 2000.00);
  Laptop dell = new Laptop("Dell XPS", 1500.00);
  Printer canon = new Printer("Canon LBP6310", 150.75);
  Printer hp = new Printer("HP Deskjet 5820", 120.00);
  Monitor thunder = new Monitor("Apple ThunderBolt", 1800.00);
  Monitor dellCurve = new Monitor("Dell Curve GT", 900.00);
  AllinOne aio1 = new AllinOne(mac, canon, thunder);
  System.out.println("It should print Apple MacBook Pro/Canon LBP6310/Apple ThunderBolt AllinOne and it prints: " + aio1.getName());
  System.out.println("It should print 3800.00 and it prints: " + aio1.getPrice());
  AllinOne aio2 = new AllinOne(dell, hp, dellCurve);
  System.out.println("It should print Dell XPS/HP Deskjet 5820/Dell Curve GT AllinOne and it prints: " + aio2.getName());
  System.out.println("It should print 2400.00 and it prints: " + aio2.getPrice());

  System.out.println("It should print 1 and it prints: " + aio1.compareTo(aio2));
  }
}
// End of main
```